

## Lab Observation

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12. Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux).

### **Aim:**

To demonstrate virtualization by installing and using a Type-2 Hypervisor (VMware Workstation) on a host operating system and creating and configuring a Virtual Machine (VM).

### **Procedure:**

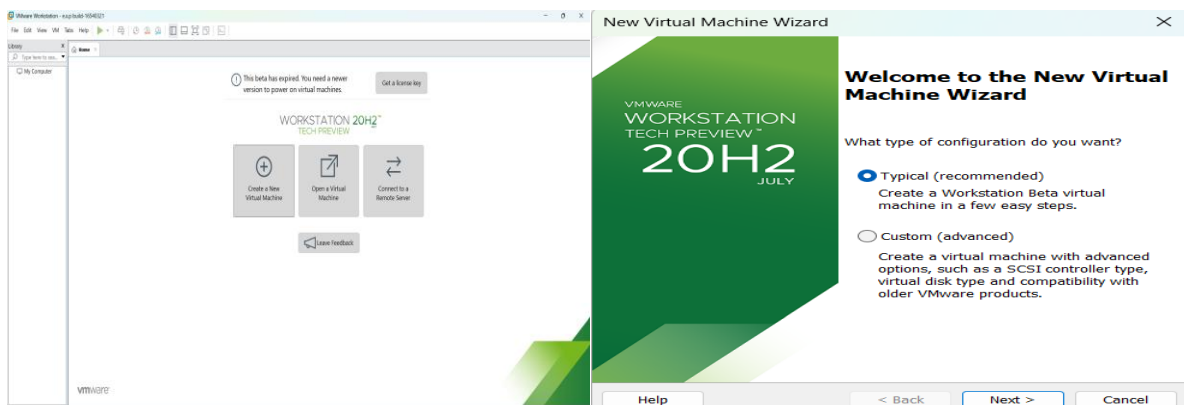
#### Step 1 – Start VMware Workstation

1. Open VMware Workstation in Windows.
2. VMware Workstation acts as the Type-2 Hypervisor.
3. Click Create a New Virtual Machine.

#### Step 2 – Create the VM

1. Select Typical (Recommended).
2. Click Next.
3. Select the Ubuntu ISO file.
4. Click Next.
5. Open Terminal and Run Comments.

### **Implementation:**



New Virtual Machine Wizard

### Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

D:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-ar Browse...

Ubuntu 64-bit 18.04.3 detected.  
This operating system will use Easy Install. [\(What's this?\)](#)

☐ I will install the operating system later.  
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

New Virtual Machine Wizard

### Easy Install Information

This is used to install Ubuntu 64-bit.

Personalize Linux

Full name: k.dhilip

User name: dhilip

Password: .....

Confirm: .....

Help < Back Next > Cancel

New Virtual Machine Wizard

### Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu 64-bit

Location:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu 64-bit Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

New Virtual Machine Wizard

### Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

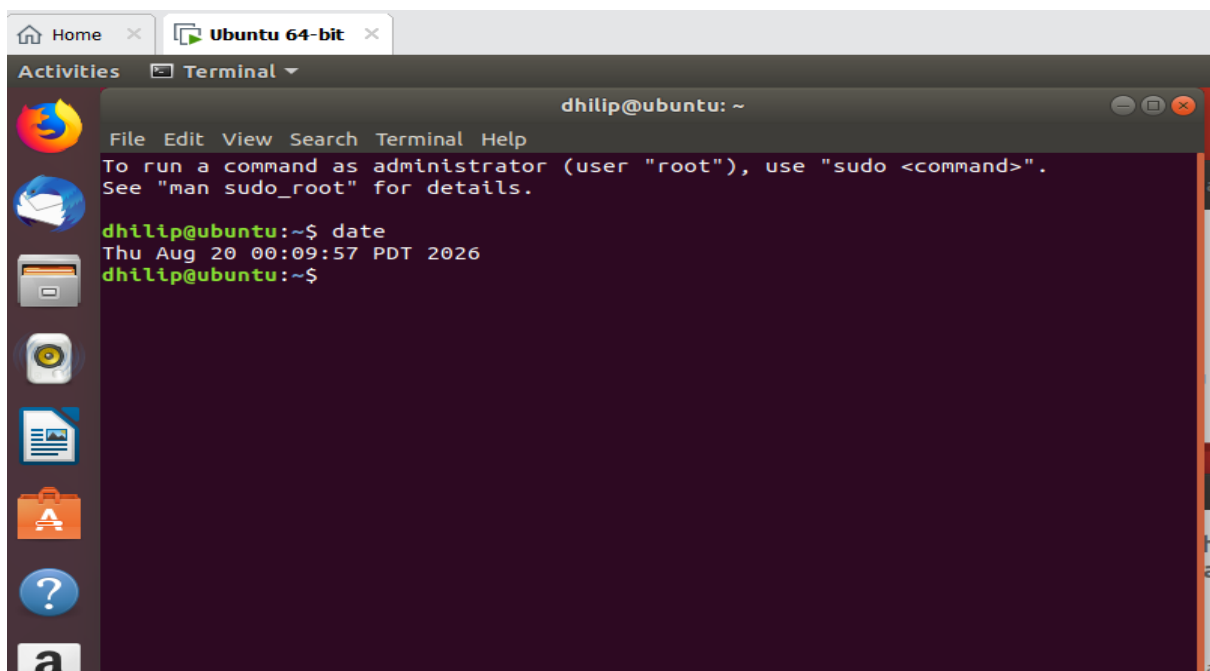
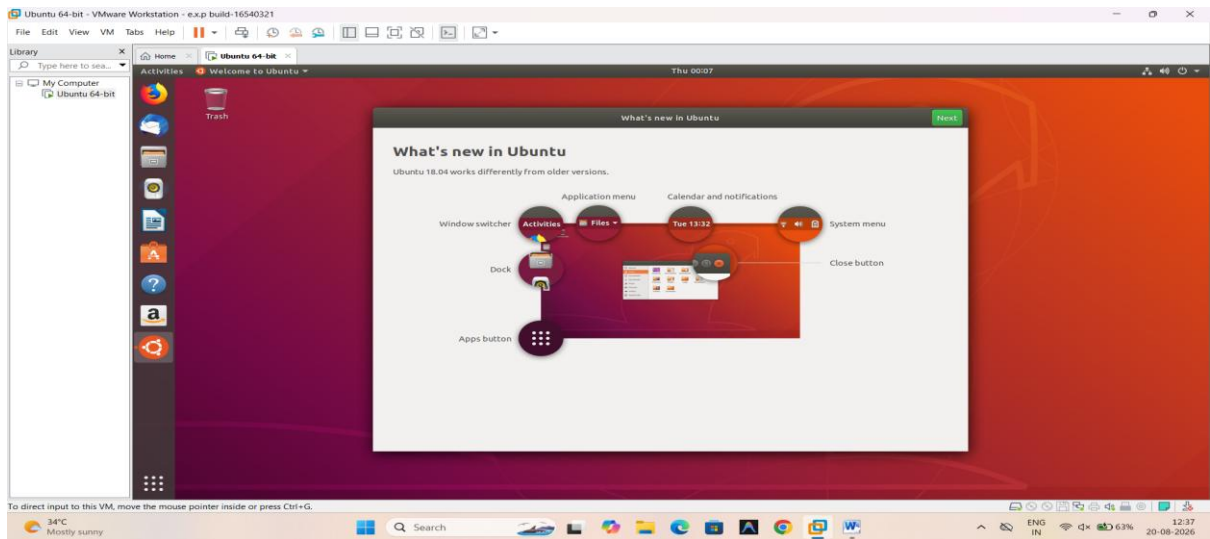
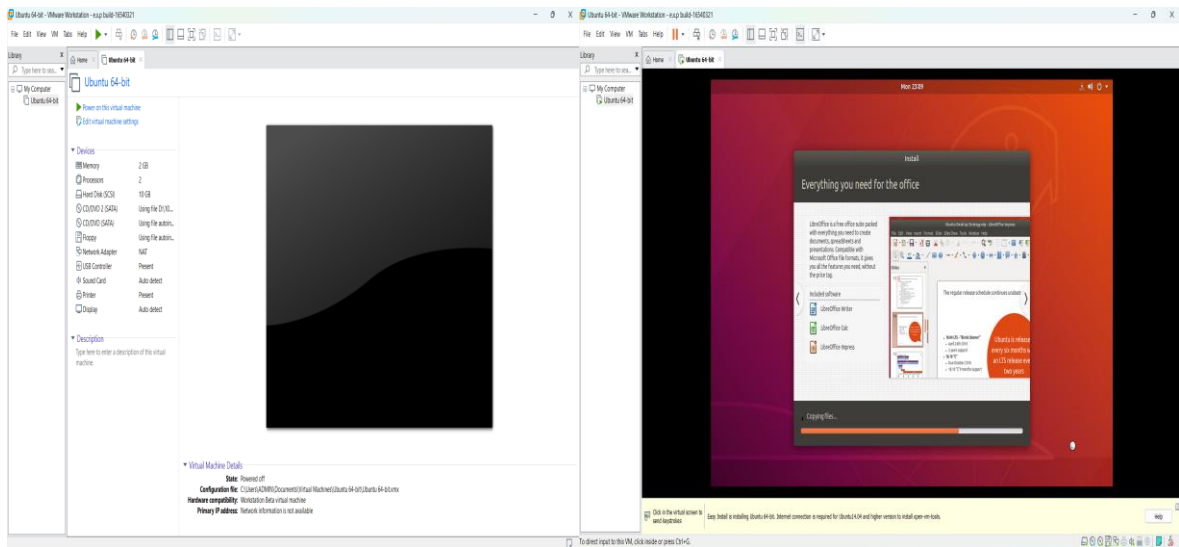
The virtual machine will be created with the following settings:

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Name:             | Ubuntu 64-bit                                            |
| Location:         | C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu 6...    |
| Version:          | Workstation Beta                                         |
| Operating System: | Ubuntu 64-bit                                            |
| Hard Disk:        | 10 GB, Split                                             |
| Memory:           | 2048 MB                                                  |
| Network Adapter:  | NAT                                                      |
| Other Devices:    | 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C... |

Customize Hardware...

☒ Power on this virtual machine after creation

< Back Finish Cancel



### **13. Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.**

#### **Aim:**

To create a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage disk using a Type 2 virtualization software (VMware Workstation).

#### **Procedure:**

##### Step 1: Open VMware

1. Open VMware Workstation.
2. Click Create a New Virtual Machine.
3. Select Typical (Recommended).
4. Click Next.

##### Step 2: Select Operating System

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select your Ubuntu ISO file.
4. Click Next.

##### Step 3: Enter VM Information

Enter:

- Full Name: Your name
- Username: Your preferred username
- Password: Your preferred password

Click Next.

If VMware does not show this screen, you can install Ubuntu normally after creating the VM.

##### Step 4: Give VM Name and Location

For example:

- Virtual Machine Name: Ubuntu-15GB
- Location: Choose where you want to store the VM.

Click Next.

##### Step 5: Set Storage

When VMware asks for disk capacity:

- Maximum disk size: 15 GB

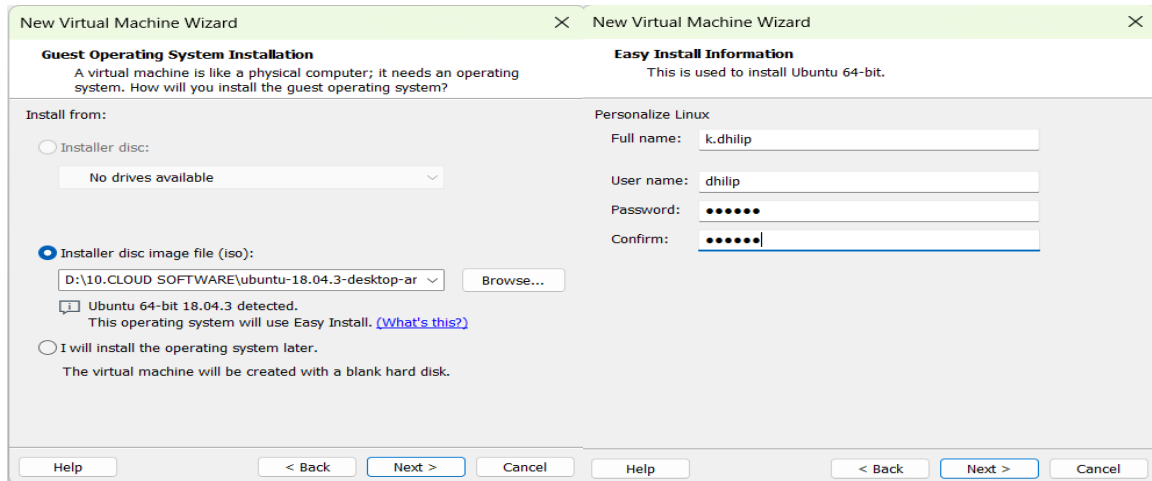
- Select Store virtual disk as a single file.

Click Next.

Step 6: Set CPU and RAM

Before finishing, click Customize Hardware.

## Implementation:



New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu-15GB

Location:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-15GB

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

15

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

Hardware

Device

Summary

Processors

2 GB

Processors

1

New CD/DVD (SATA)

Using file D:\18.CLOUD SOF...

Network Adapter

NAT

USB Controller

Present

Sound Card

Auto detect

Printer

Present

Display

Auto detect

Add...

Remove

Processors

Number of processors:

1

Number of cores per processor:

1

Total processor cores:

1

Virtualization engine

☐ Virtualize Intel VT-x/EPT or AMD-V/RVI

☐ Virtualize CPU performance counters

☐ Virtualize IOMMU (I/O memory management unit)

Close

Help

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

Name:

Ubuntu-15GB

Location:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-1...

Version:

Workstation Beta

Operating System:

Ubuntu 64-bit

Hard Disk:

15 GB

Memory:

2048 MB

Network Adapter:

NAT

Other Devices:

CD/DVD, USB Controller, Printer, Sound Card

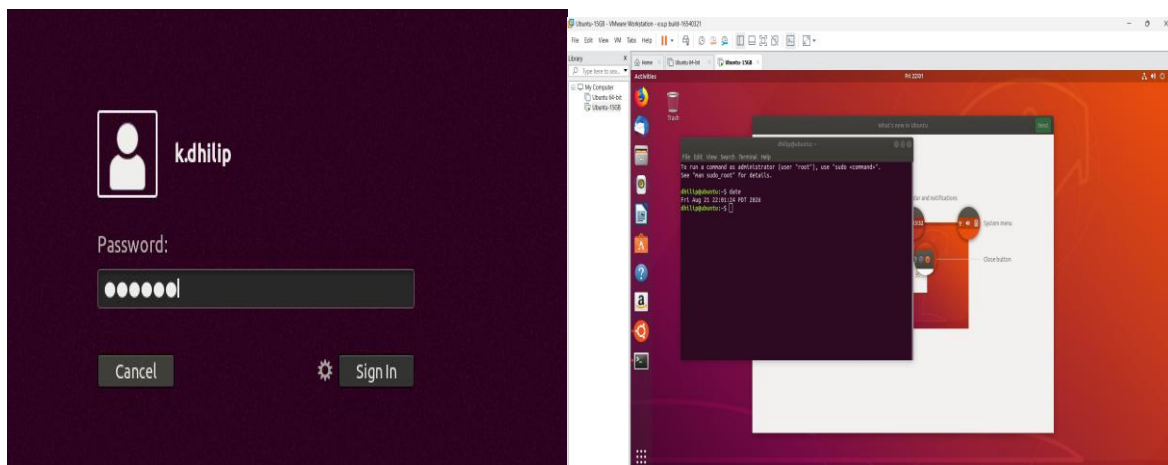
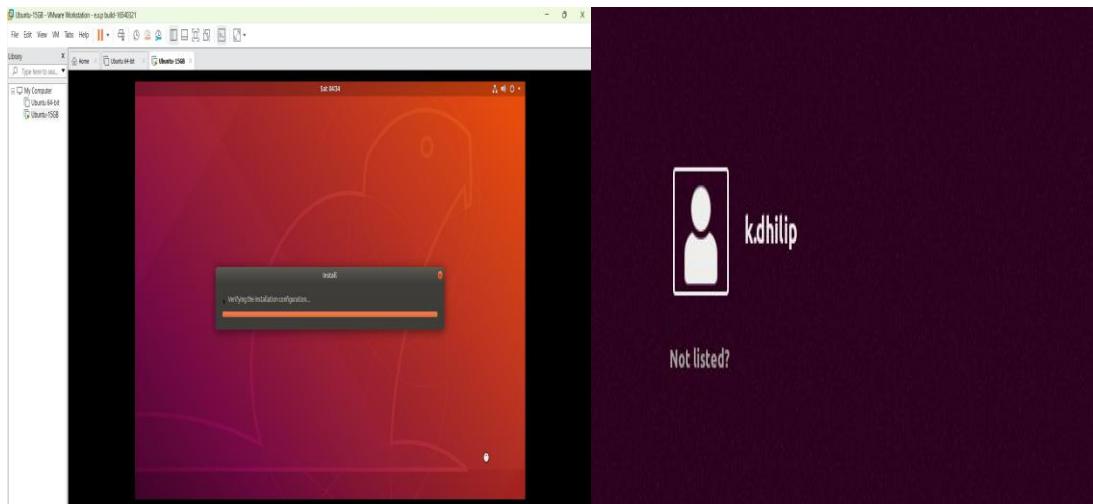
Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel



## 14. Create a Virtual Hard Disk and allocate the storage using VM ware Workstation

### Aim:

To create a Virtual Hard Disk and allocate storage to it using VMware Workstation.

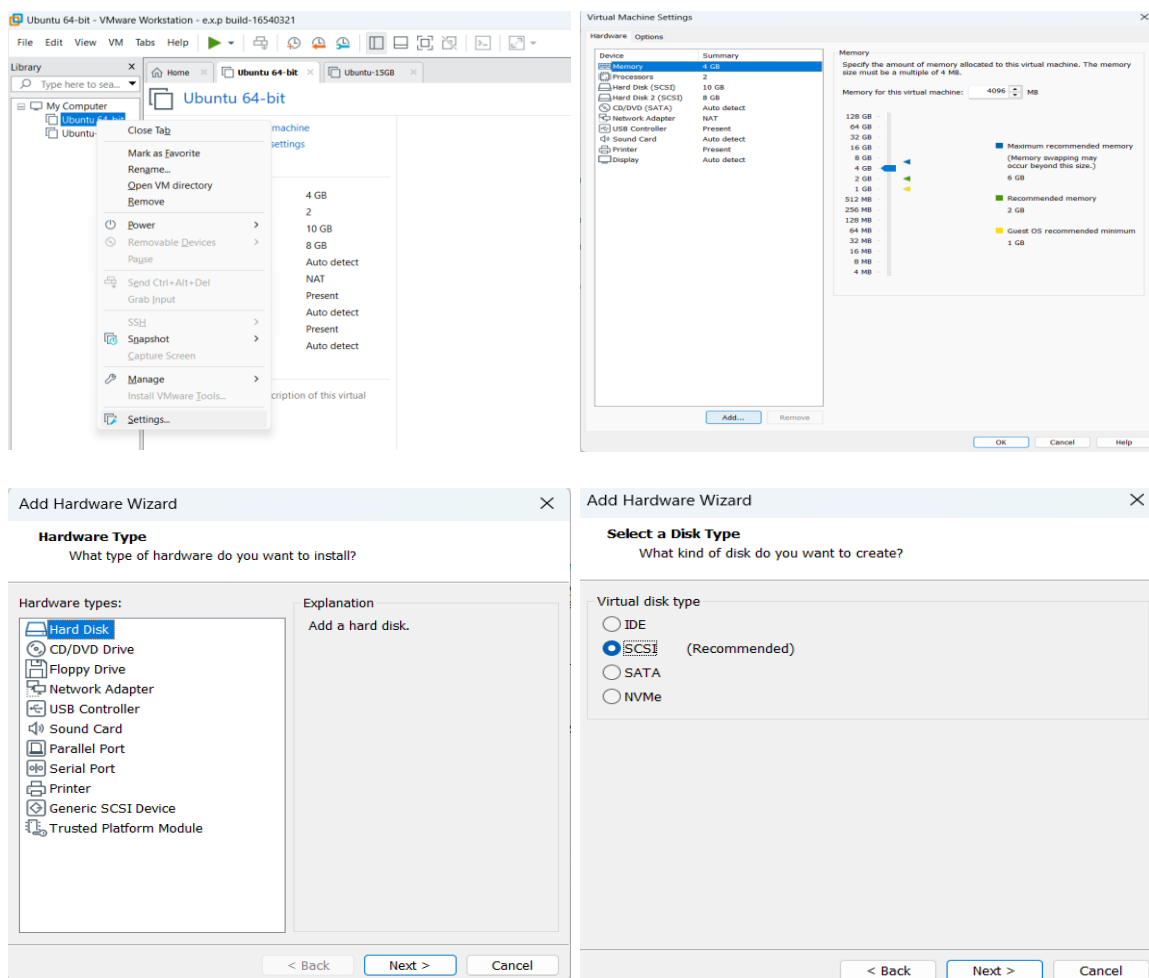
### Procedure:

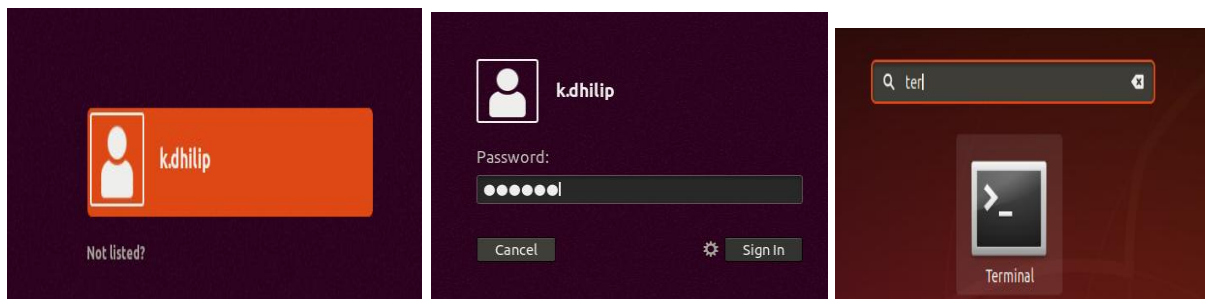
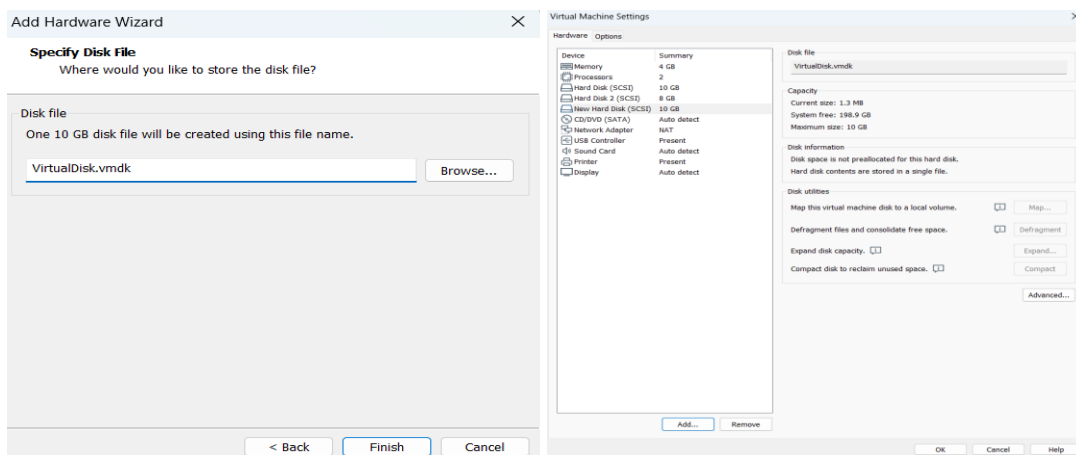
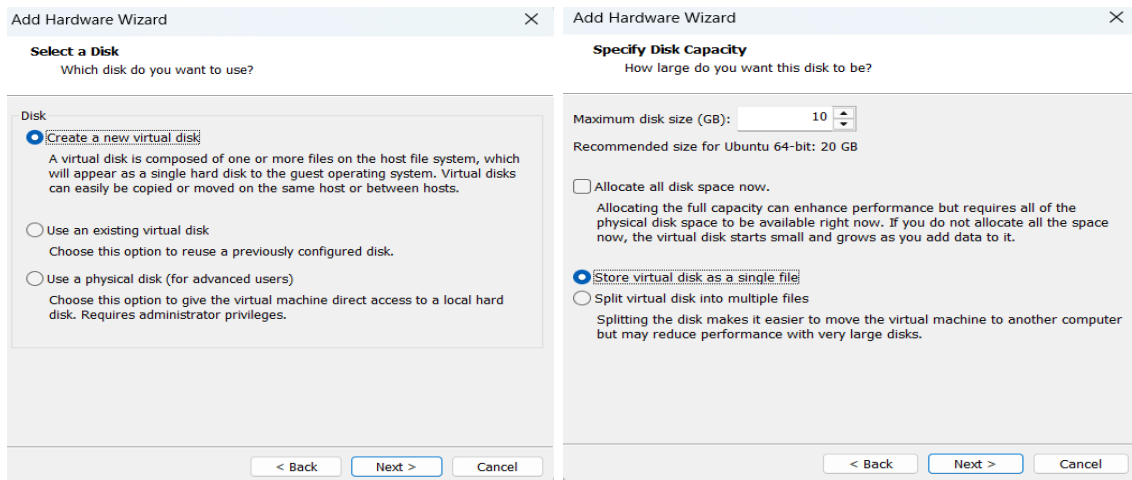
1. Open VMware Workstation.
2. Select your existing Ubuntu VM.
3. Make sure the VM is powered off.
4. Right-click the VM and select Settings.
5. Click Add... at the bottom of the Hardware tab.
6. Select Hard Disk.
7. Click Next.
8. Select the disk type:  
SCSI (recommended/default).
9. Click Next.

10. Select:  
Create a new virtual disk.
11. Click Next.
12. Enter the required disk size.
13. Select:
14. Click Next.
15. Give the virtual disk a name, for example:
16. Click Finish.
17. Click OK to save the VM settings.

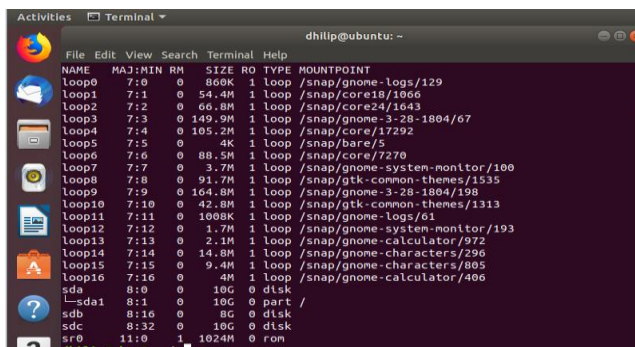
Start the Ubuntu VM and open Terminal and run the comments.

## Implementation:





**Command: lsblk**



## 15. Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM

### Aim:

To create a snapshot of a Virtual Machine using VMware Workstation and test it by restoring the VM to the previous saved state.

### Procedure:

Step 1: Start the VM

1. Open VMware Workstation.
2. Select your Ubuntu VM.
3. Start the VM.
4. Wait until Ubuntu loads completely.

Step 2: Create a Snapshot

1. In VMware Workstation, select the running Ubuntu VM.
2. Go to:

VM → Snapshot → Take Snapshot

3. Enter a name:

Before\_Test

4. You can enter a description such as:

Snapshot before making changes

5. Click Take Snapshot.

The current state of the VM is now saved.

Step 3: Make a Change

Open Ubuntu Terminal and create a test file:

```
touch test_snapshot.txt
```

Check it:

```
ls
```

You should see:

```
test_snapshot.txt
```

#### Step 4: Restore the Previous Snapshot

Now restore the VM to the snapshot you created earlier.

1. In VMware Workstation, go to:

VM → Snapshot → Snapshot Manager

2. Select:

Before\_Test

3. Click Go To.
4. Confirm the restore operation if VMware asks.

VMware will return the VM to the state when the snapshot was taken.

#### Step 5: Test the Snapshot

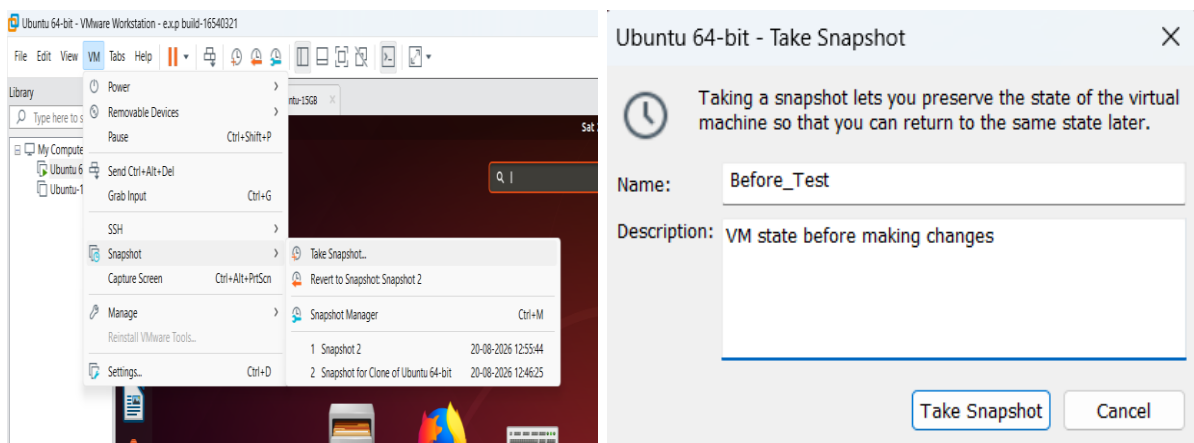
After Ubuntu starts again, open Terminal and run:

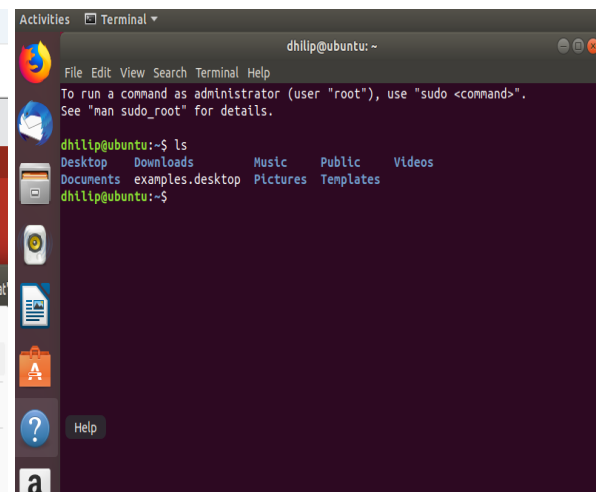
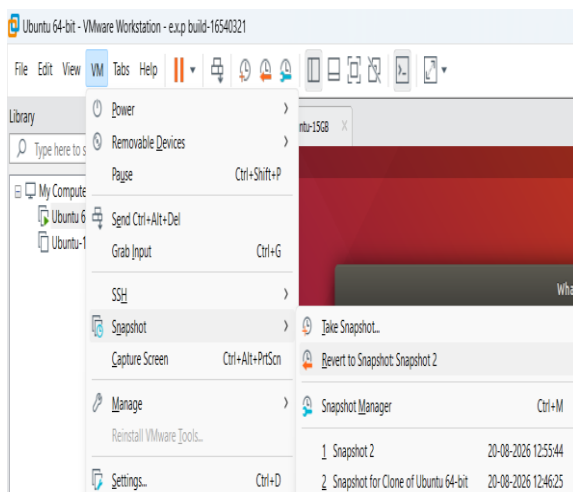
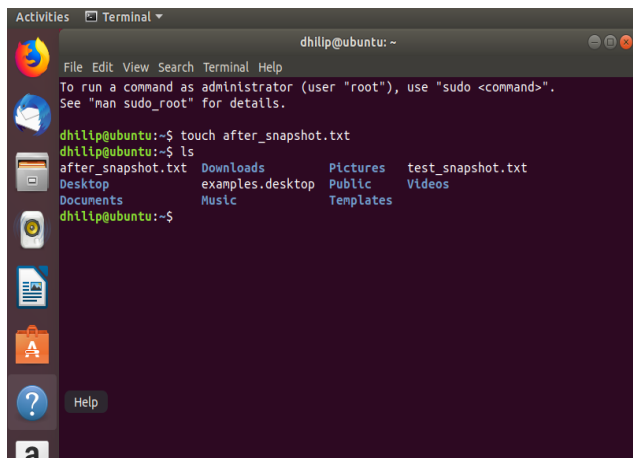
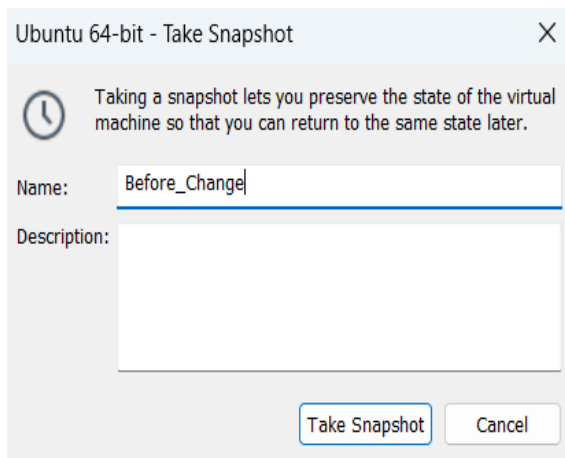
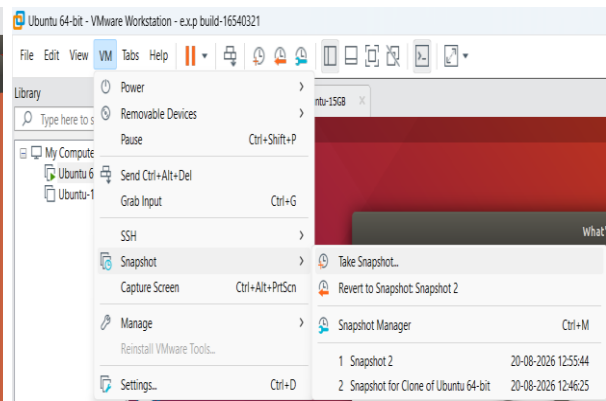
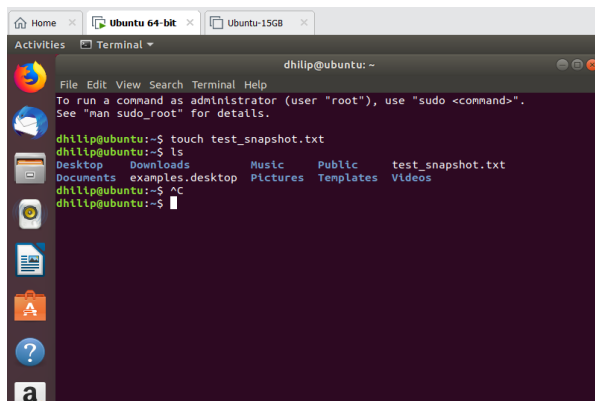
ls

The file:

test\_snapshot.txt

#### Implementation:





## 16. Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

### Aim:

To create a clone of an existing Virtual Machine using VMware Workstation and test the cloned VM by loading and running it successfully.

### Requirements:

- VMware Workstation
- An existing Ubuntu/Linux VM
- Sufficient storage space for the cloned VM

### Procedure:

#### Step 1: Power Off the Original VM

1. Open VMware Workstation.
2. Select your existing Ubuntu VM.
3. Make sure the VM is powered off.
  - If it is running, shut down Ubuntu normally.

#### Step 2: Create the Clone

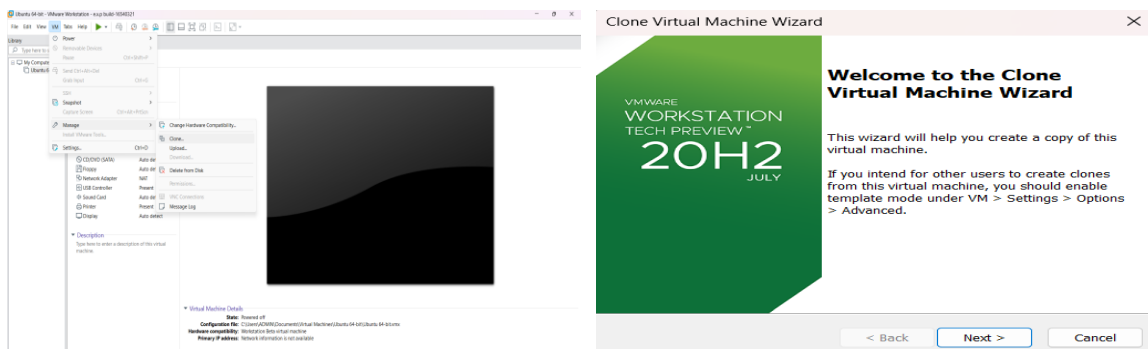
1. Select the original VM.
2. Go to:

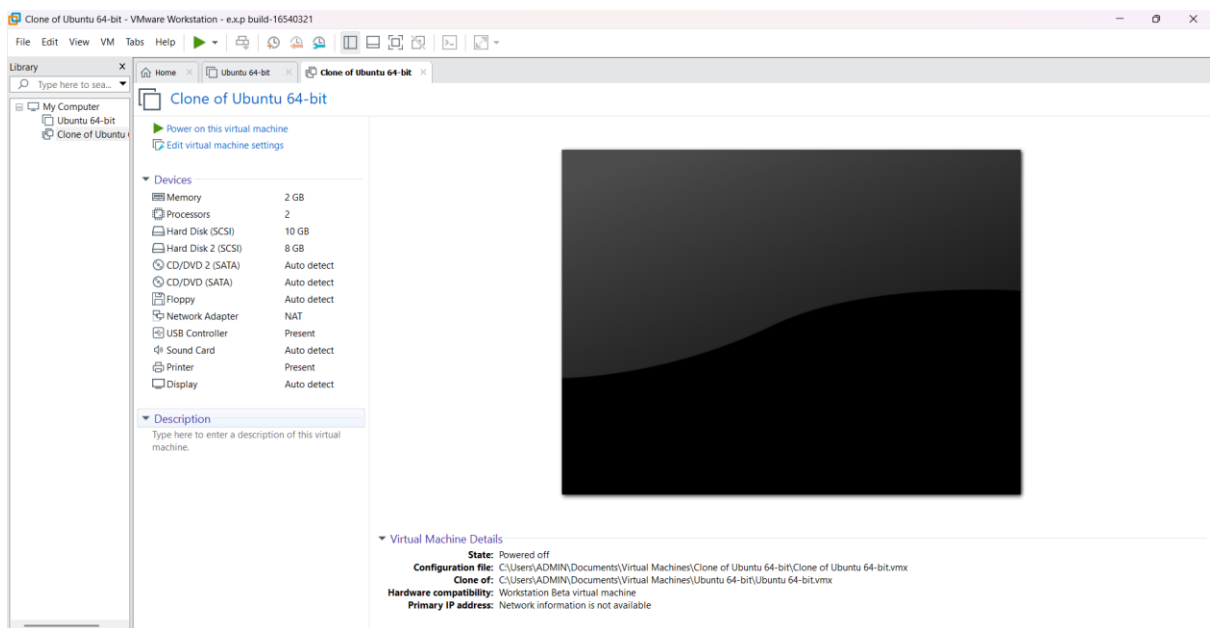
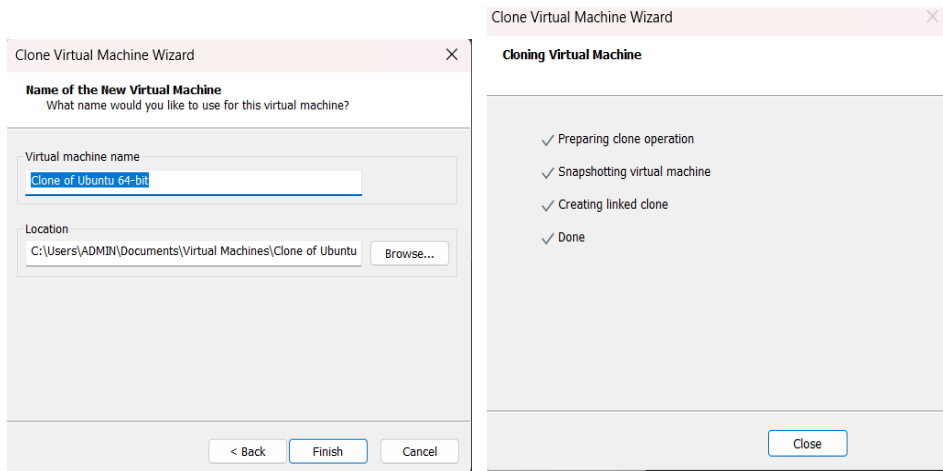
VM → Manage → Clone

3. The Clone Virtual Machine Wizard will open.
4. Click Next.
5. Select:

The current state in the virtual machine

6. Click Next.





## **17.Change Hardware compatibility of a VM (Either by clone/create new one) which is already created and configured and run simple unix commands**

### **Aim:**

To change the hardware compatibility of an already created and configured virtual machine in VMware Workstation and verify the VM by running simple UNIX commands.

### **Procedure:**

#### Step 1: Open VMware Workstation

1. Open VMware Workstation in Windows.
2. Select your already created Ubuntu VM.
3. Do not start the VM.
4. Make sure the Ubuntu VM is powered off.

#### Step 2: Open Virtual Machine Settings

1. Select the Ubuntu VM.
2. Click VM from the top menu.
3. Select Manage.
4. Click Change Hardware Compatibility.

#### Step 3: Select Hardware Compatibility

1. A Change Hardware Compatibility Wizard will open.
2. Click Next.
3. VMware will display the current hardware compatibility version.
4. Select a newer hardware compatibility version available in your VMware Workstation.
5. Click Next.

Example: If your current version is older, select the latest compatible version offered by your VMware Workstation.

#### Step 4: Choose the Conversion Option

1. VMware may ask whether to:
  - Create a new virtual machine, or
  - Change the existing virtual machine.
2. Select the appropriate option for the existing VM.
3. Click Next.

#### Step 5: Complete the Hardware Compatibility Change

1. Review the settings.
2. Click Finish.
3. Wait until VMware completes the hardware compatibility change.

- The VM's hardware compatibility has now been changed.

## Step 6: Start the Ubuntu VM

1. Select the modified Ubuntu VM.
2. Click Power on.
3. Wait for Ubuntu to boot completely.

## Step 7: Open Terminal

Open the Ubuntu Terminal using:

Ctrl + Alt + T

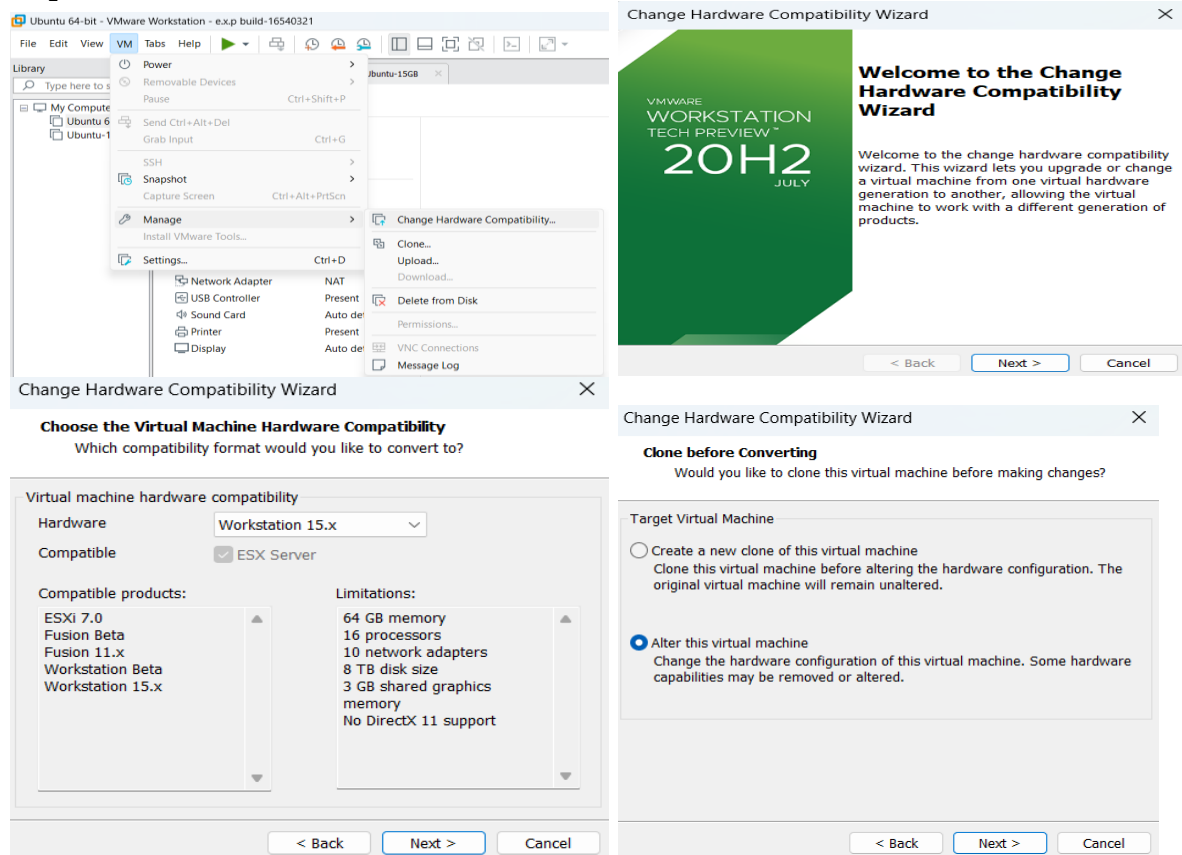
## Step 8: Run Simple UNIX Commands

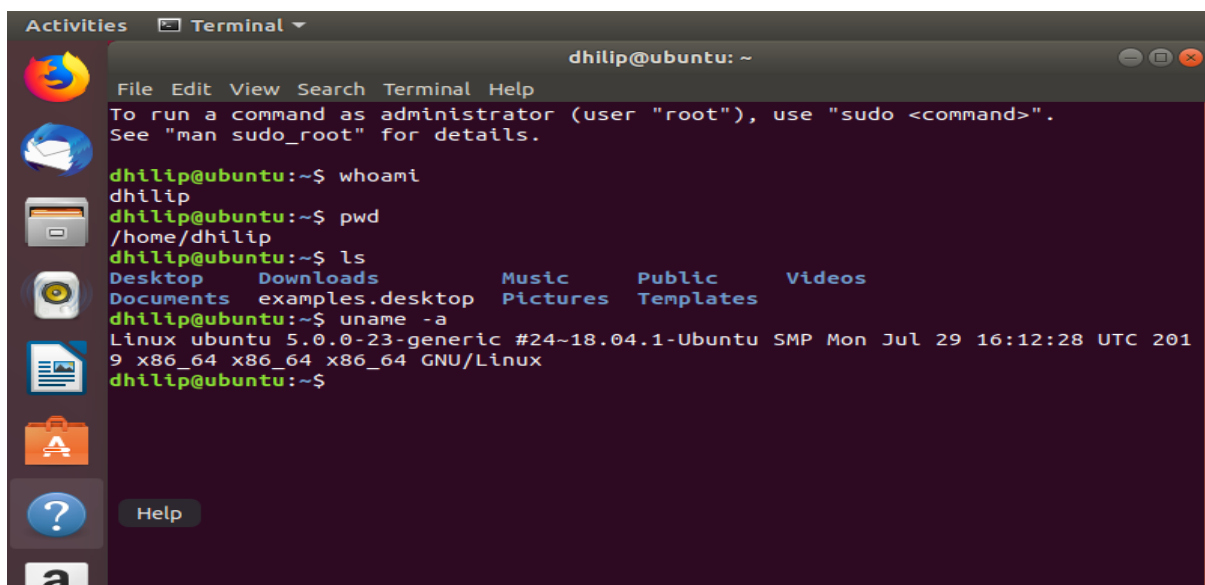
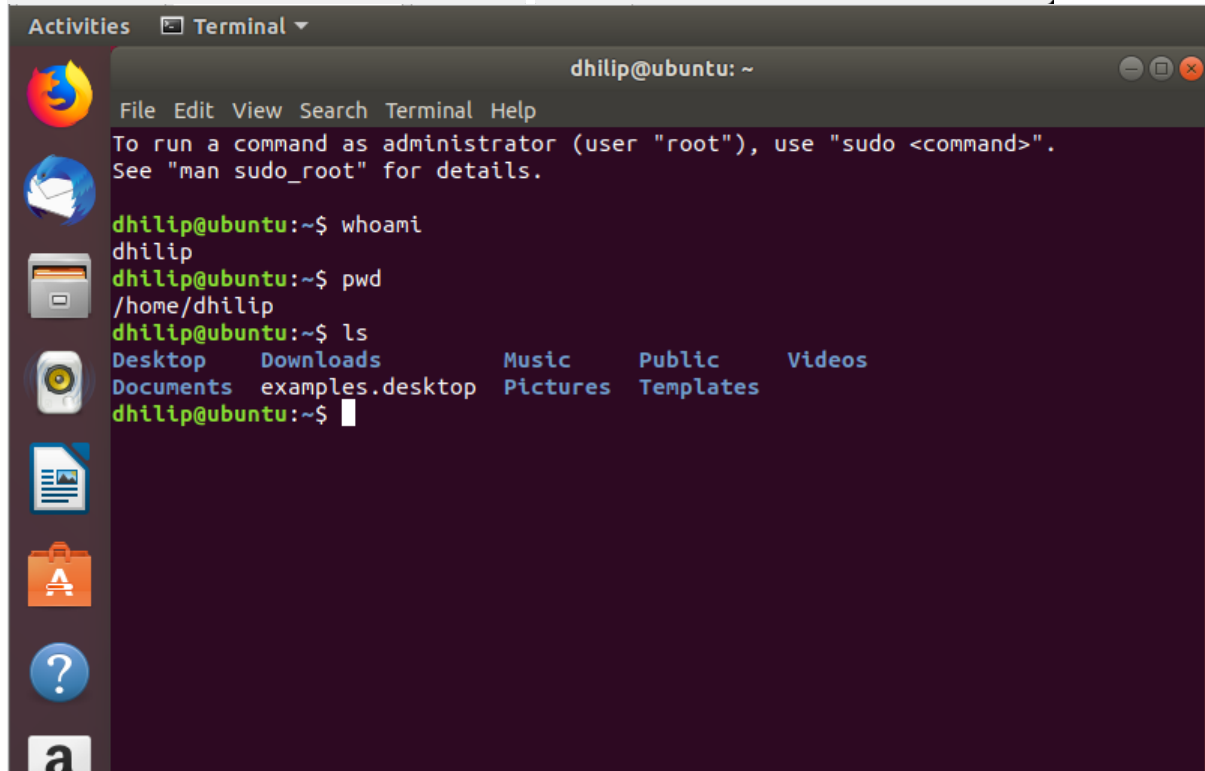
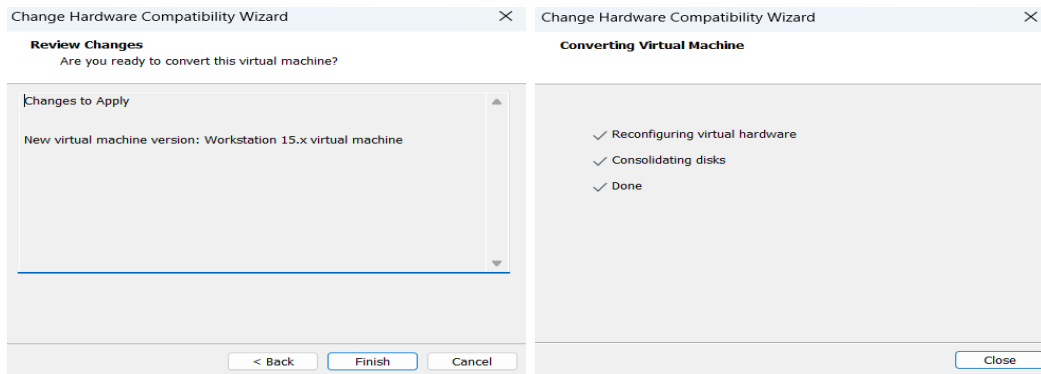
Run the following commands one by one.

1. Display current user:

Whoami

## Implementation:





## **18. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.**

### **Aim:**

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

### **Procedure:**

Step 1: Open VMware Workstation

1. Open VMware Workstation in Windows.
2. Click Create a New Virtual Machine.

Step 2: Select Installation Method

1. Select Typical (recommended).
2. Click Next.

Step 3: Select the Operating System ISO

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select the Linux/Windows ISO file.
4. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

1. Select the appropriate operating system.
2. For Ubuntu, select:

Linux

Ubuntu 64-bit

3. Click Next.

Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

Click Next.

#### Step 6: Configure Disk

1. Specify the disk size.

For example:

20 GB

2. Select Store virtual disk as a single file.
3. Click Next.

#### Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

#### Step 8: Install the Operating System

1. Start the virtual machine by clicking Power on this virtual machine.
2. The operating system installer will start.
3. Follow the installation instructions.
4. Create a username and password when requested.
5. Wait for the installation to complete.

#### Step 9: Restart the Virtual Machine

After installation:

1. Restart the VM.
2. Remove the ISO if VMware asks you to do so.
3. Ubuntu will boot from the virtual hard disk.

#### Step 10: Verify the Installation

Open the Terminal in Ubuntu and run:

```
uname -a
```

Then:

```
ls
```

And

Pwd

## Implementation:



WORKSTATION 20H2™  
TECH PREVIEW JULY

Create a New Virtual Machine

Open a Virtual Machine

Connect to a Remote Server

Leave Feedback



New Virtual Machine Wizard

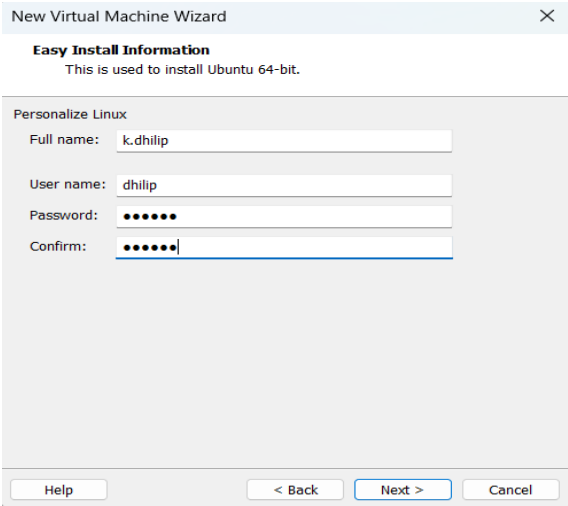
Welcome to the New Virtual Machine Wizard

What type of configuration do you want?

☒ Typical (recommended)  
Create a Workstation Beta virtual machine in a few easy steps.

☐ Custom (advanced)  
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.

Help < Back Next > Cancel



New Virtual Machine Wizard

**Easy Install Information**  
This is used to install Ubuntu 64-bit.

Personalize Linux

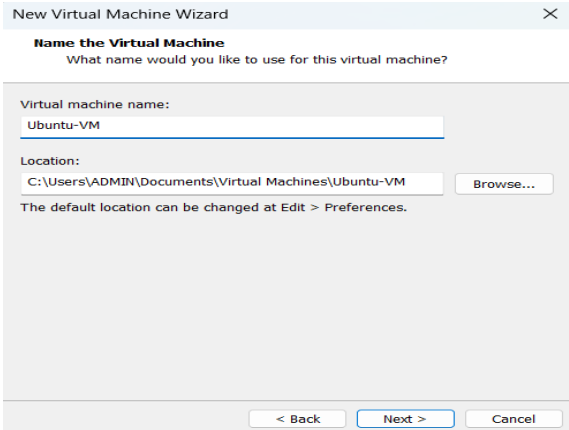
Full name: k.dhilip

User name: dhilip

Password: •••••

Confirm: •••••

Help < Back Next > Cancel



New Virtual Machine Wizard

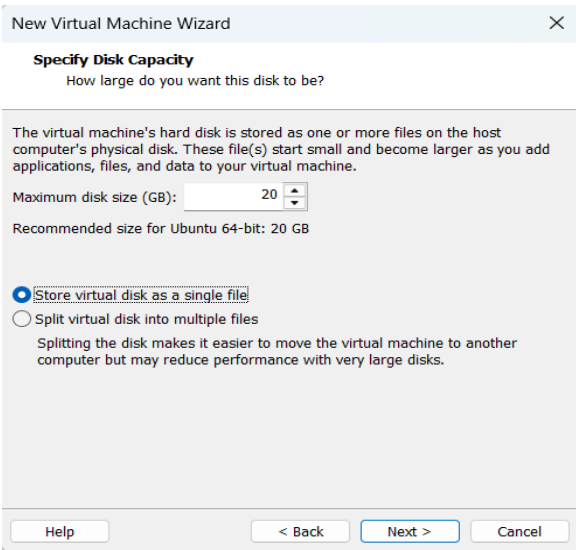
**Name the Virtual Machine**  
What name would you like to use for this virtual machine?

Virtual machine name:  
Ubuntu-VM

Location:  
C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel



New Virtual Machine Wizard

**Specify Disk Capacity**  
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 20

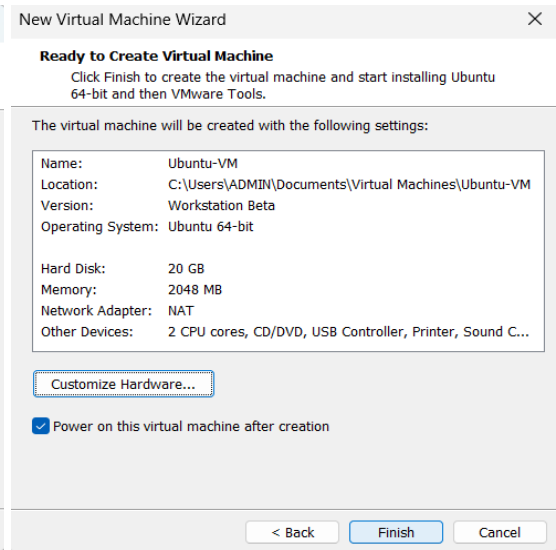
Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel



New Virtual Machine Wizard

**Ready to Create Virtual Machine**  
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

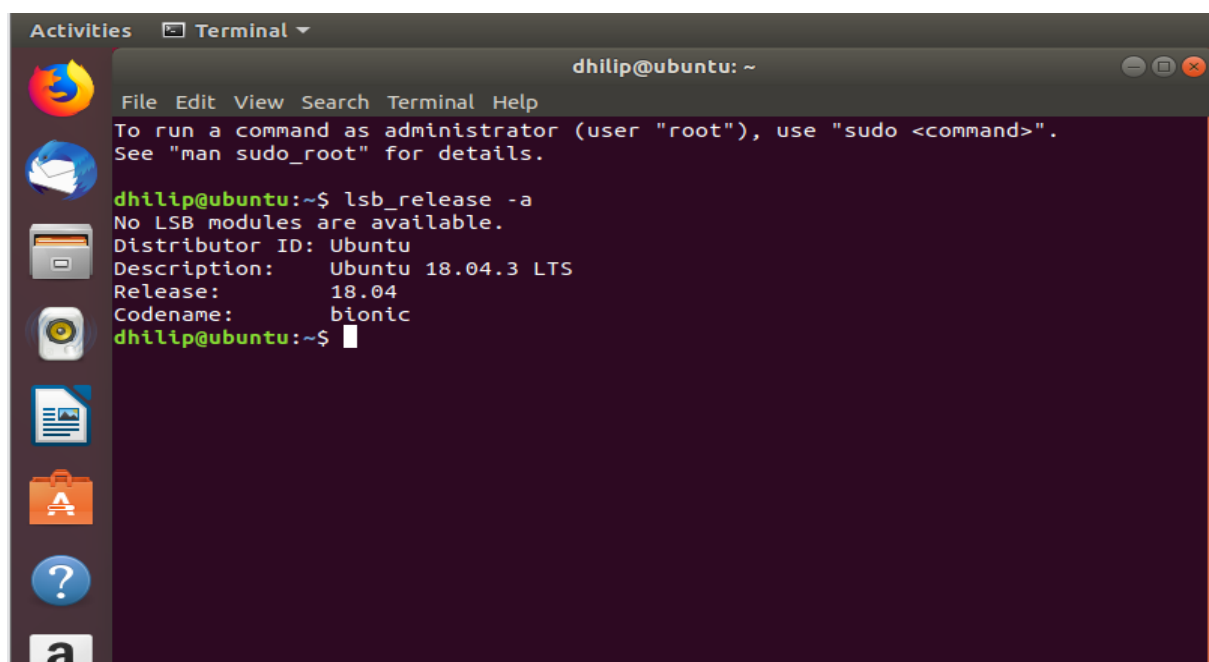
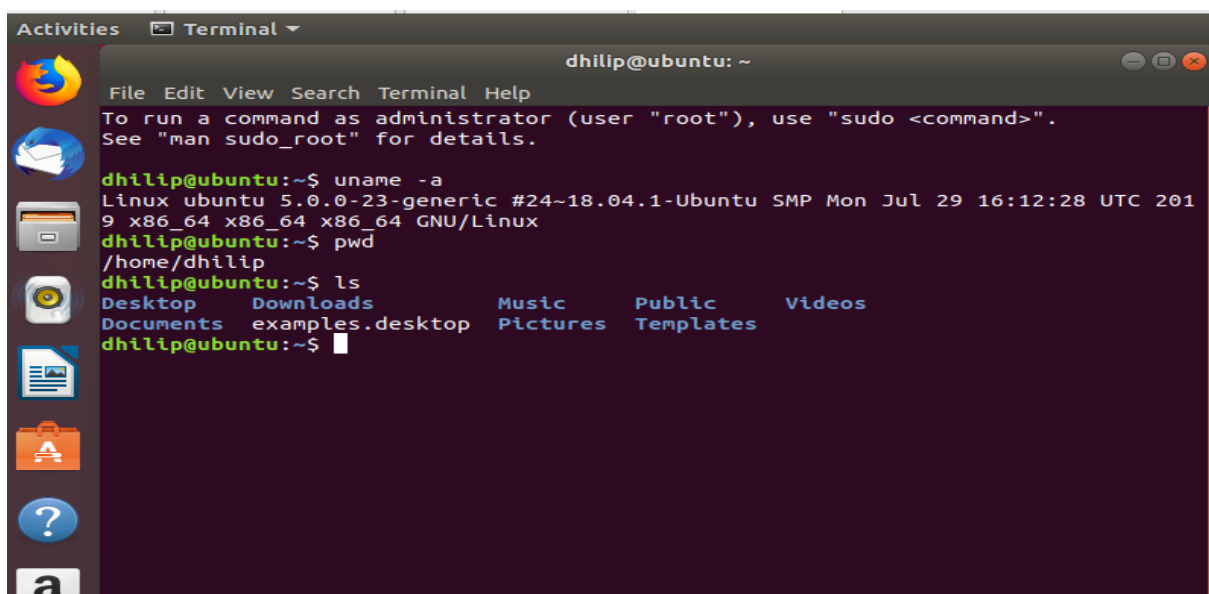
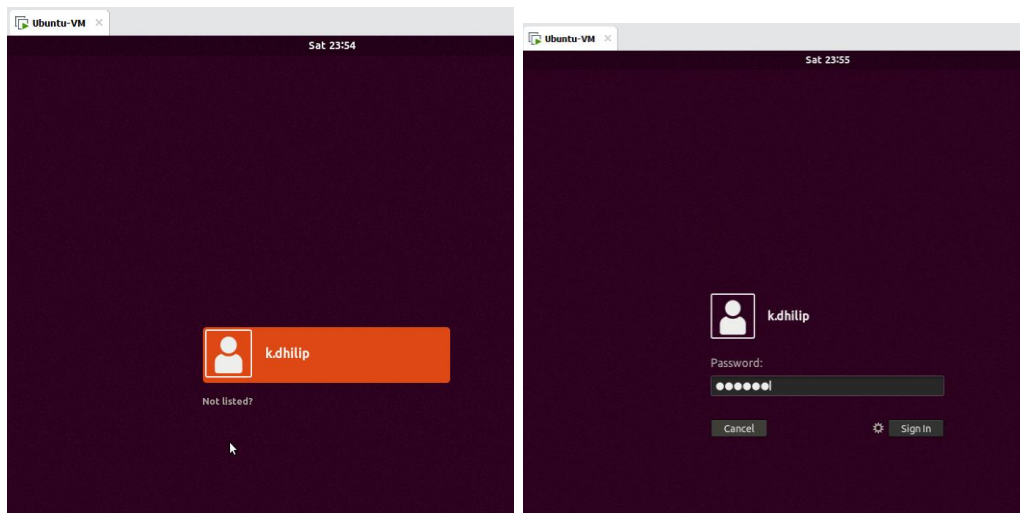
The virtual machine will be created with the following settings:

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Name:             | Ubuntu-VM                                                |
| Location:         | C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM      |
| Version:          | Workstation Beta                                         |
| Operating System: | Ubuntu 64-bit                                            |
| Hard Disk:        | 20 GB                                                    |
| Memory:           | 2048 MB                                                  |
| Network Adapter:  | NAT                                                      |
| Other Devices:    | 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C... |

Customize Hardware...

☒ Power on this virtual machine after creation

< Back Finish Cancel



## **19. Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.**

### **Aim:**

To install a C compiler (GCC) in the Ubuntu virtual machine created using VMware Workstation and execute a simple C program.

### **Procedure:**

Step 1: Start Ubuntu VM

1. Open VMware Workstation.
2. Start your Ubuntu VM.
3. Login to Ubuntu.

Step 2: Open Terminal

Press:

Ctrl + Alt + T

Step 3: Update the Package List

Type:

`sudo apt update`

Press Enter.

Enter your Ubuntu password when asked.

While typing the password, you won't see any characters. This is normal. Type the password and press Enter.

Wait until the update finishes.

Step 4: Install GCC Compiler

Type:

`sudo apt install gcc`

If it asks:

Do you want to continue? [Y/n]

Type:

Y

and press Enter.

Wait for the installation to complete.

#### Step 5: Check GCC Installation

Type:

```
gcc --version
```

You should see something similar to:

```
gcc (Ubuntu ...) ...
```

This confirms that the C compiler is installed.

#### Step 6: Create a C Program

Create a file named hello.c:

```
nano hello.c
```

The Nano editor will open.

Type this program:

```
#include <stdio.h>
```

```
int main()  
{  
    printf("Hello World!\n");  
    return 0;  
}
```

#### Step 7: Save the Program

In Nano:

1. Press Ctrl + O
2. Press Enter
3. Press Ctrl + X

You will return to the terminal.

#### Step 8: Compile the C Program

Type:

```
gcc hello.c -o hello
```

If there is no error message, compilation was successful.

Step 9: Execute the Program

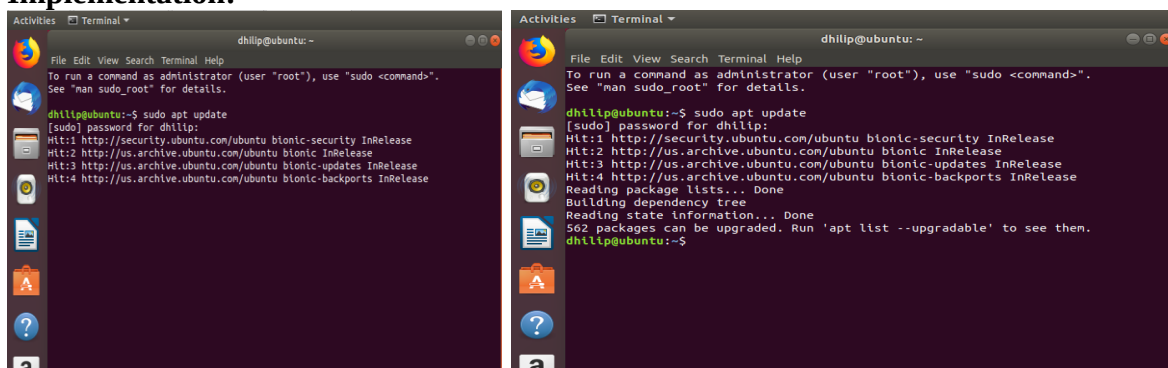
Type:

```
./hello
```

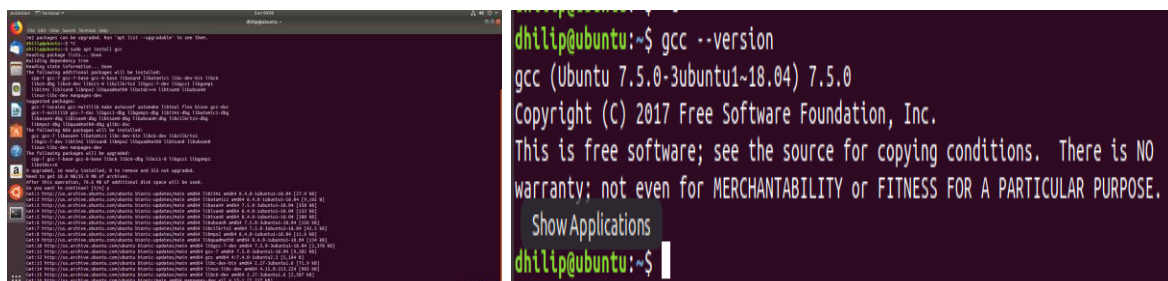
Expected output:

Hello World!

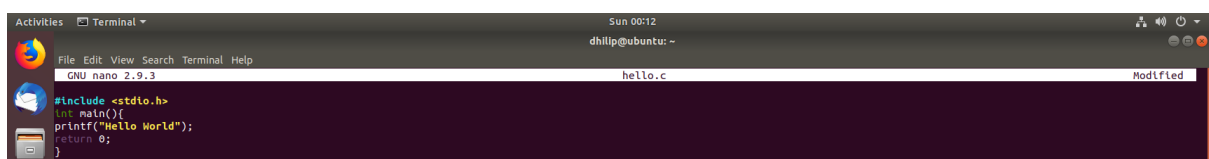
## Implementation:



The first terminal screenshot shows the command `dhilip@ubuntu:~$ sudo apt update` being executed. It prompts for a password and then displays the output of the update process, including hits for various security and update packages. The second terminal screenshot shows the same command being executed again, with the same output.

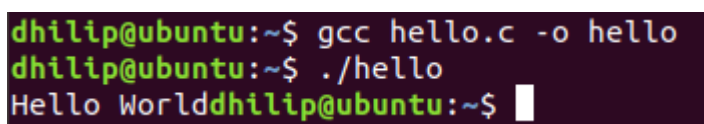


The first terminal screenshot shows the command `dhilip@ubuntu:~$ gcc --version` being executed. It displays the version of the GCC compiler (7.5.0) and the copyright information. The second terminal screenshot shows the same command being executed again, with the same output.



The terminal screenshot shows the command `dhilip@ubuntu:~$ nano hello.c` being executed. It opens the nano text editor with a file named `hello.c`. The content of the file is as follows:

```
#include <stdio.h>
int main()
{
    printf("Hello World!");
    return 0;
}
```



The terminal screenshot shows the command `dhilip@ubuntu:~$ gcc hello.c -o hello` being executed, followed by `dhilip@ubuntu:~$ ./hello`. The output of the program is `Hello World!`.

## **20. Install Virtualbox with different flavours of linux or windows OS on top of windows 10 using custom installation.**

### **Aim:**

To install Oracle VirtualBox on Windows 10 and install a different flavour of Linux/Windows operating system using Custom Installation.

### **Procedure:**

#### Step 1: Open VirtualBox

1. Open Oracle VM VirtualBox on your Windows computer.
2. Click New to create a new virtual machine.

#### Step 2: Create the Virtual Machine

1. Enter the VM name, for example:

Ubuntu-VirtualBox

2. Select the location where you want to store the VM.
3. Select the ISO Image of Ubuntu.
4. Select the operating system type:
  - Type: Linux
  - Version: Ubuntu (64-bit)
5. Click Next.

#### Step 3: Select Custom Installation

When VirtualBox provides the installation options, choose the option for custom/unattended installation settings as applicable.

If you see Skip Unattended Installation, select it if you want to perform the OS installation manually.

Click Next.

#### Step 4: Configure Hardware

Set the hardware resources:

Memory:

2048 MB (2 GB)

Processors:

2 CPUs

Click Next.

#### Step 5: Create Virtual Hard Disk

1. Select Create a Virtual Hard Disk Now.
2. Set the disk size to:

20 GB

3. Click Next.
4. Review the configuration.
5. Click Finish.

#### Step 6: Start the Virtual Machine

1. Select Ubuntu-VirtualBox.
2. Click Start.
3. Ubuntu will boot from the selected ISO file.

#### Step 7: Install Ubuntu

When the Ubuntu installer appears:

1. Select Install Ubuntu.
2. Select the keyboard layout.
3. Choose the installation type.
4. Select Erase disk and install Ubuntu.

This refers to the virtual hard disk, not your actual Windows C: drive.

5. Set your location/time zone.
6. Create a username and password.
7. Click Install.
8. Wait for the installation to complete.

#### Step 8: Restart the VM

After installation:

1. Click Restart Now.
2. If VirtualBox asks you to remove the installation medium, press Enter.
3. Ubuntu will restart.
4. Login using the username and password you created.

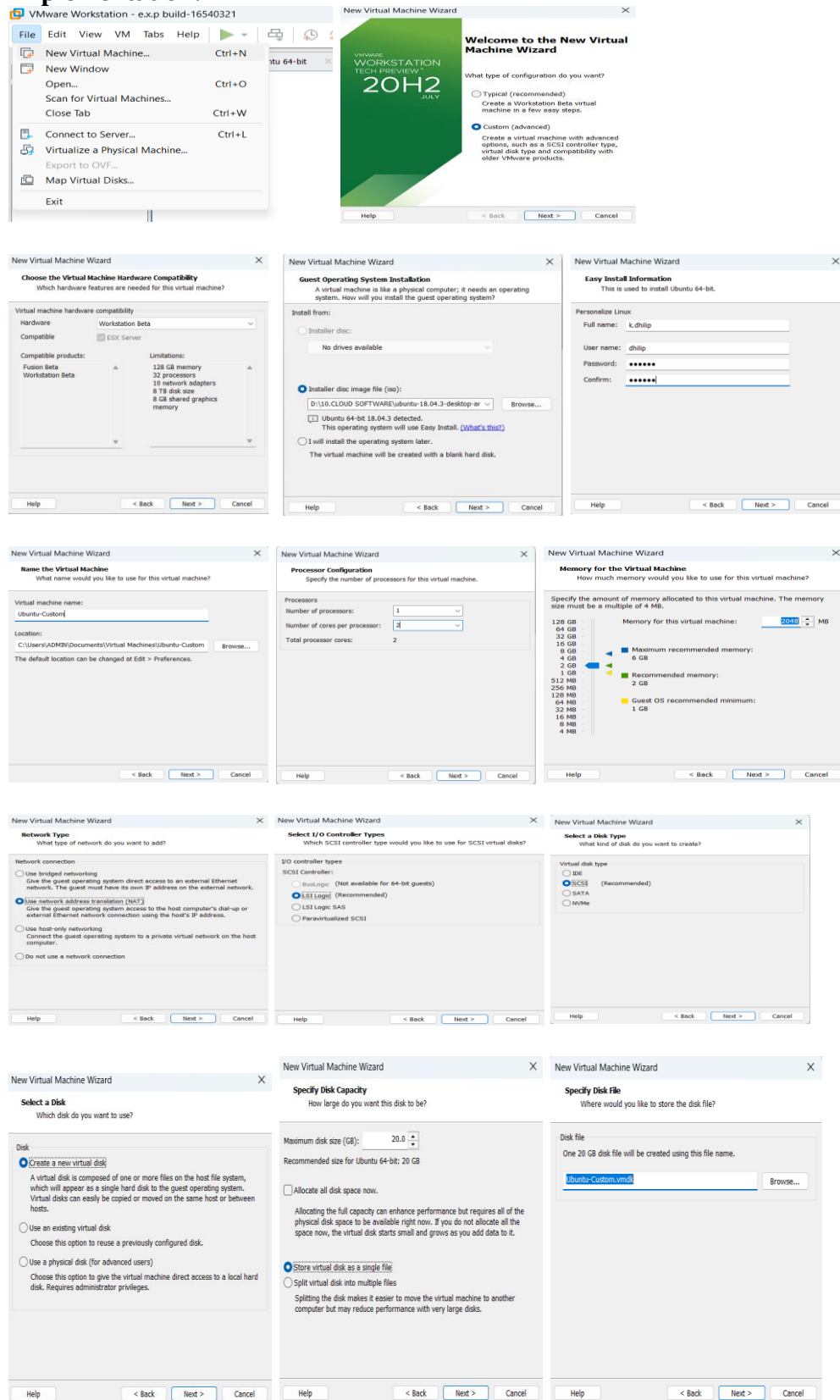
#### Step 9: Verify the Installation

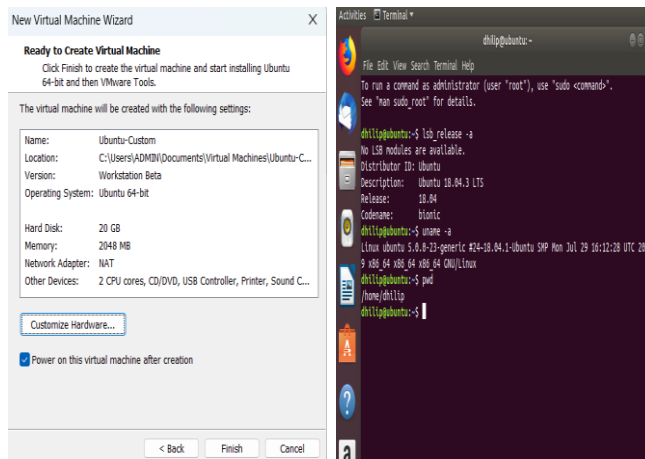
Open Terminal:

Run:

lsb\_release -a

## Implementation:





## 21. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.

### Aim:

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

### Procedure:

Step 1: Open VMware Workstation

3. Open VMware Workstation in Windows.
4. Click Create a New Virtual Machine.

Step 2: Select Installation Method

3. Select Typical (recommended).
4. Click Next.

Step 3: Select the Operating System ISO

5. Select Installer disc image file (ISO).
6. Click Browse.
7. Select the Linux/Windows ISO file.
8. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

3. Select the appropriate operating system.
4. For Ubuntu, select:

Linux  
Ubuntu 64-bit

4. Click Next.

#### Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

Click Next.

#### Step 6: Configure Disk

2. Specify the disk size.

For example:

20 GB

4. Select Store virtual disk as a single file.
5. Click Next.

#### Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

#### Step 8: Install the Operating System

6. Start the virtual machine by clicking Power on this virtual machine.
7. The operating system installer will start.
8. Follow the installation instructions.
9. Create a username and password when requested.
10. Wait for the installation to complete.

#### Step 9: Restart the Virtual Machine

After installation:

4. Restart the VM.
5. Remove the ISO if VMware asks you to do so.
6. Ubuntu will boot from the virtual hard disk.

## Step 10: Verify the Installation

Open the Terminal in Ubuntu and run:

```
uname -a
```

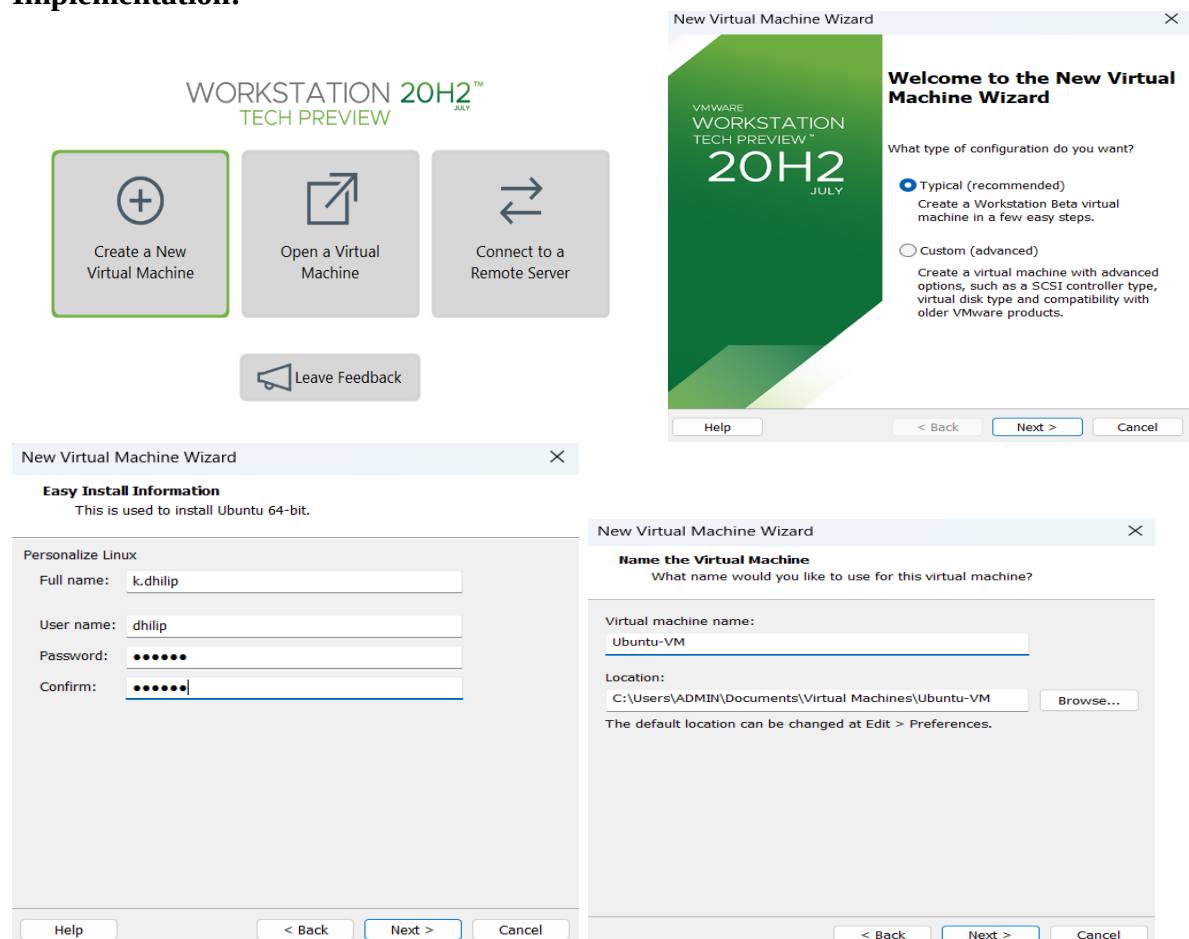
Then:

```
ls
```

And:

```
Pwd
```

## Implementation:



New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.

The virtual machine will be created with the following settings:

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Name:             | Ubuntu-VM                                                |
| Location:         | C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM      |
| Version:          | Workstation Beta                                         |
| Operating System: | Ubuntu 64-bit                                            |
| Hard Disk:        | 20 GB                                                    |
| Memory:           | 2048 MB                                                  |
| Network Adapter:  | NAT                                                      |
| Other Devices:    | 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C... |

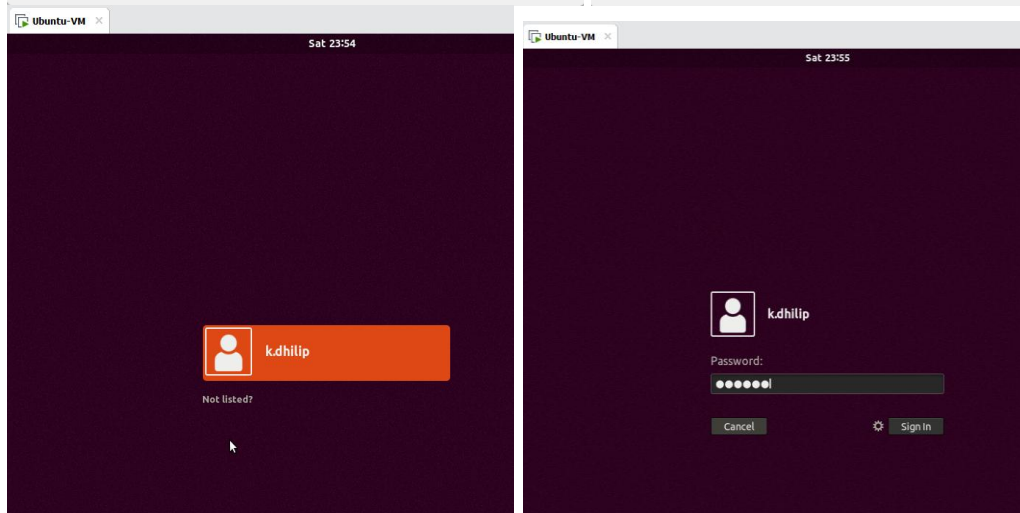
Customize Hardware...

☒ Power on this virtual machine after creation

< Back

Finish

Cancel



Activities Terminal

dhilip@ubuntu: ~

File Edit View Search Terminal Help

To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo\_root" for details.

dhilip@ubuntu:~\$ uname -a

Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 2019 x86\_64 x86\_64 x86\_64 GNU/Linux

dhilip@ubuntu:~\$ pwd

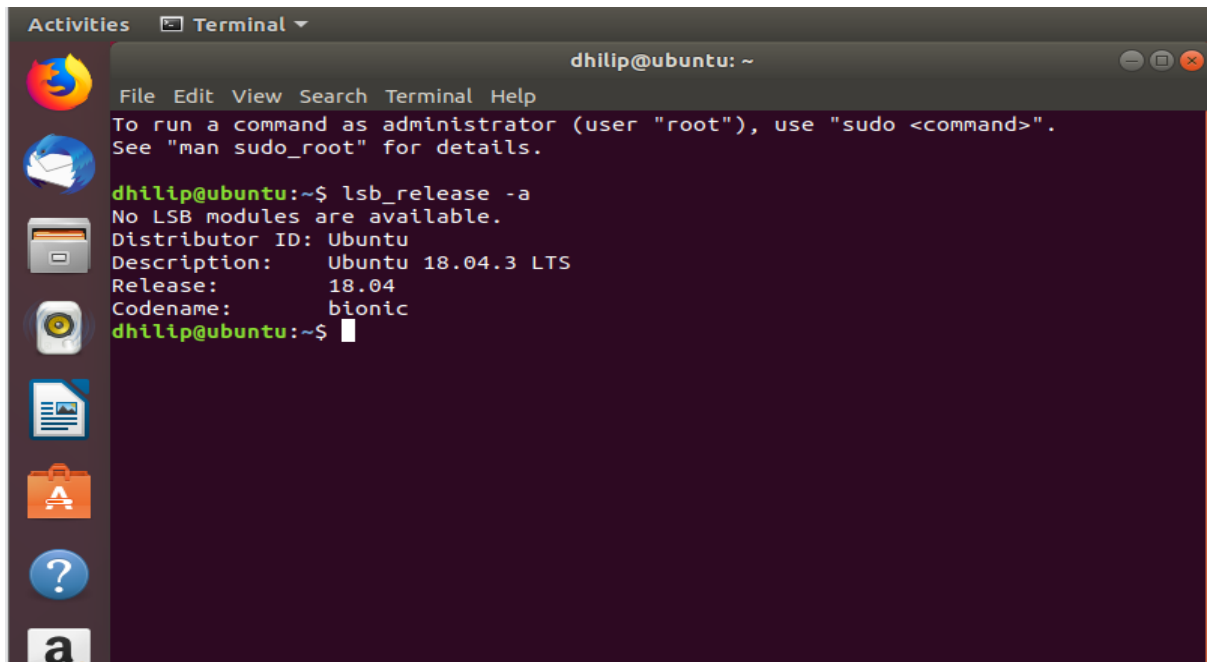
/home/dhilip

dhilip@ubuntu:~\$ ls

Desktop Downloads Music Public Videos

Documents examples.desktop Pictures Templates

dhilip@ubuntu:~\$

A screenshot of a terminal window titled 'Terminal' with a dark background. The window shows the command 'lsb\_release -a' being executed. The output displays system information: 'No LSB modules are available.', 'Distributor ID: Ubuntu', 'Description: Ubuntu 18.04.3 LTS', 'Release: 18.04', and 'Codename: bionic'. The prompt 'dhilip@ubuntu:~\$' is visible at the end of the command line.

```
dhilip@ubuntu: ~  
File Edit View Search Terminal Help  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
dhilip@ubuntu:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 18.04.3 LTS  
Release:       18.04  
Codename:      bionic  
dhilip@ubuntu:~$
```

**22. To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.**

**Aim:**

To run virtual machines with different hardware configurations in VMware Workstation and determine how many virtual machines can be utilized simultaneously on the host computer.

**Procedure:**

Step 1 — Open VMware Workstation

1. Open VMware Workstation Pro.
2. Make sure your previously created Ubuntu VMs are available.

Step 2 — Check the Host Computer Resources

Before starting multiple VMs, check your Windows computer's:

- RAM
- CPU
- Available disk space

You can check RAM and CPU using:

Task Manager → Performance

### Step 3 — Configure the First VM

For example, configure Ubuntu VM 1 with:

RAM: 2 GB

CPU: 1 processor, 2 cores

Start the VM.

### Step 4 — Configure the Second VM

Create/select another VM and configure it with:

RAM: 2 GB

CPU: 1 processor, 2 cores

Start the second VM without shutting down the first VM.

### Step 5 — Configure Additional VMs

If your computer has sufficient resources, start another VM.

For example:

VM 1 → 2 GB RAM

VM 2 → 2 GB RAM

VM 3 → 2 GB RAM

Do not start more VMs if the host system becomes very slow or runs out of memory.

### Step 6 — Check Running Virtual Machines

In VMware Workstation, observe which VMs are powered on.

You can also check Windows:

Task Manager → Performance → Memory / CPU

Observe the resource usage while the VMs are running.

### Step 7 — Test Each VM

Open the Terminal in each running Ubuntu VM and execute:

```
uname -a
```

and:

```
pwd
```

If the commands execute successfully, the VMs are running correctly.

## Step 8 — Determine the Maximum Number

Start the VMs one by one.

After starting each VM, check:

- CPU usage
- RAM usage
- System responsiveness
- Whether the VM operates normally

The maximum number of VMs is the number that can run simultaneously while the host and VMs remain usable.

## Example

If your computer has 8 GB RAM, you might test:

VM 1 → 2 GB

VM 2 → 2 GB

VM 3 → 2 GB

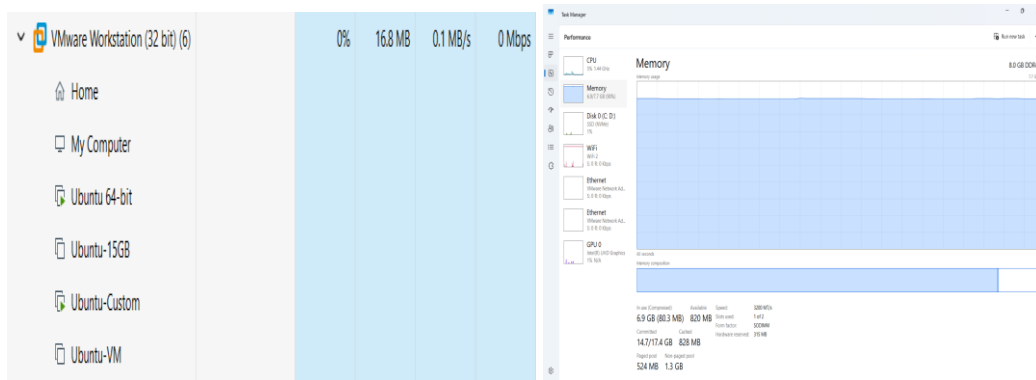
## Implementation:

The screenshot displays the VMware Workstation interface. The main window shows the 'Virtual Machine Settings' for 'Ubuntu 64-bit'. The 'Memory' tab is selected, showing a summary of 2 GB memory and 2 processors. The 'Memory' section on the right allows adjusting the memory size, with a slider set to 2048 MB. Below the slider, a bar chart shows memory usage for various VMs: Ubuntu 64-bit (2 GB), Ubuntu-15GB (15 GB), Ubuntu-VM (2 GB), and Ubuntu-Custom (1 GB). The 'Guest OS recommended minimum' is 1 GB, and the 'Maximum recommended memory' is 6 GB.

In the bottom-left corner, a terminal window is open, showing the command `uname -a` and its output: `Linux ubuntu 5.0.0-23-generic #24-18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux`.

In the bottom-right corner, a 'My Computer' window shows a table of VMs and their status:

| Name          | Status      |
|---------------|-------------|
| Ubuntu 64-bit | Powered on  |
| Ubuntu-15GB   | Powered off |
| Ubuntu-Custom | Powered on  |
| Ubuntu-VM     | Powered off |



## 23. Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

### Aim:

To attach a virtual block (virtual hard disk) to a virtual machine and verify whether the data stored on it remains available even after the virtual machine is released or powered off.

### Procedure:

We will use your Ubuntu VM in VMware Workstation Pro.

#### Step 1 — Power Off the VM

1. Open VMware Workstation Pro.
2. Select your Ubuntu VM.
3. Shut down Ubuntu completely.
4. Make sure the VM shows Powered Off.

Do not suspend it. It should be completely powered off.

#### Step 2 — Open Virtual Machine Settings

1. Select the Ubuntu VM.
2. Click Edit virtual machine settings.
3. The Virtual Machine Settings window will open.

#### Step 3 — Add a Virtual Hard Disk

1. Click Add...
2. Select:

#### Hard Disk

3. Click Next.

#### Step 4 — Select Disk Type

Select:

SCSI

Then click Next.

#### Step 5 — Create a New Virtual Disk

Select:

Create a new virtual disk

Click Next.

#### Step 6 — Set Disk Size

Enter:

1 GB

You can use 1 GB because we only need a small disk for this experiment.

Select:

Store virtual disk as a single file

Click Next.

#### Step 7 — Finish Adding the Disk

Leave the disk filename as the default.

Click:

Finish

Then click:

OK

The new virtual hard disk is now attached to your Ubuntu VM.

#### Step 8 — Start Ubuntu

Power on the Ubuntu VM.

Open Terminal:

Ctrl + Alt + T

## Step 9 — Check the New Virtual Disk

Run:

```
lsblk
```

You should see your original disk and a new disk, for example:

```
sda
├── sda1
└── sda2
sdb
```

The newly added disk will usually appear as sdb.

Do not format sda. That is your Ubuntu system disk.

## Step 10 — Create a Partition

If the new disk is /dev/sdb, run:

```
sudo fdisk /dev/sdb
```

Inside fdisk:

Press:

```
n
```

Press Enter.

Then press Enter for the default partition options until the partition is created.

Finally press:

```
w
```

This saves the partition.

## Step 11 — Format the Partition

Run:

```
sudo mkfs.ext4 /dev/sdb1
```

This creates an ext4 filesystem on the new virtual disk.

## Step 12 — Create a Mount Directory

Run:

```
sudo mkdir /mnt/virtualdisk
```

### Step 13 — Mount the Virtual Disk

Run:

```
sudo mount /dev/sdb1 /mnt/virtualdisk
```

Check:

```
df -h
```

You should see /mnt/virtualdisk.

### Step 14 — Create Data on the Virtual Block

Create a file:

```
sudo touch /mnt/virtualdisk/test_data.txt
```

Put some text into it:

```
echo "Virtual Block Test Data" | sudo tee /mnt/virtualdisk/test_data.txt
```

Check it:

```
cat /mnt/virtualdisk/test_data.txt
```

Expected:

Virtual Block Test Data

Data has now been stored on the additional virtual disk.

### Step 15 — Release/Power Off the VM

1. Shut down Ubuntu:

```
sudo shutdown -h now
```

2. Wait until VMware shows the VM as Powered Off.

### Step 16 — Start the VM Again

Power on the same Ubuntu VM.

Open Terminal.

Check the disk:

lsblk

You should see the additional disk again.

Step 17 — Mount the Virtual Disk Again

Run:

```
sudo mount /dev/sdb1 /mnt/virtualdisk
```

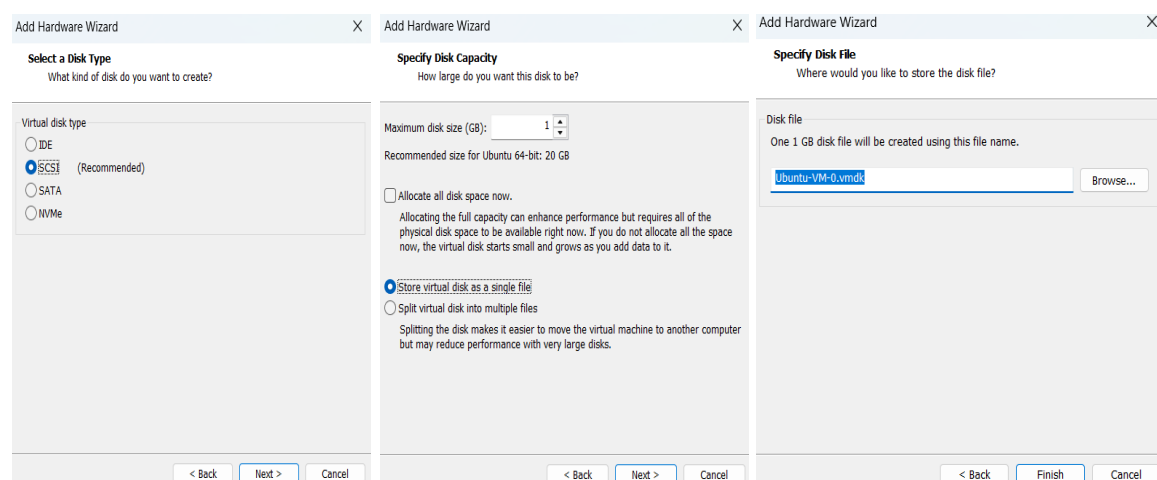
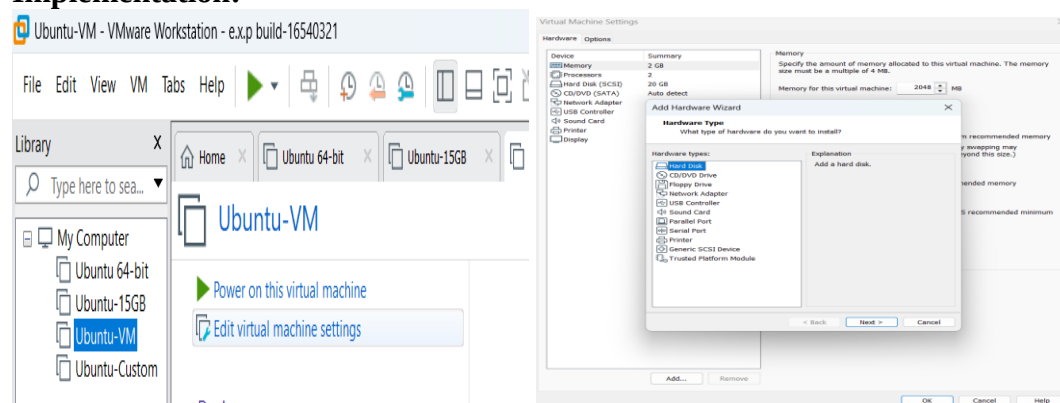
Then:

```
cat /mnt/virtualdisk/test_data.txt
```

Expected Output

Virtual Block Test Data

## Implementation:



▼ **Devices**

|                    |             |
|--------------------|-------------|
| Memory             | 2 GB        |
| Processors         | 2           |
| Hard Disk (SCSI)   | 20 GB       |
| Hard Disk 2 (SCSI) | 1 GB        |
| CD/DVD (SATA)      | Auto detect |
| Network Adapter    | NAT         |
| USB Controller     | Present     |
| Sound Card         | Auto detect |
| Printer            | Present     |
| Display            | Auto detect |

▼ **Description**

Type here to enter a description of this virtual machine.



```
dhilip@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
loop0 7:0 0 1000K 1 loop /snap/gnome-logs/61
loop1 7:1 0 54.4M 1 loop /snap/core18/1066
loop2 7:2 0 14.8M 1 loop /snap/gnome-characters/296
loop3 7:3 0 3.7M 1 loop /snap/gnome-system-monitor/100
loop4 7:4 0 88.5M 1 loop /snap/core/7270
loop5 7:5 0 149.9M 1 loop /snap/gnome-3-28-1804/67
loop6 7:6 0 4M 1 loop /snap/gnome-calculator/486
loop7 7:7 0 42.8M 1 loop /snap/gtk-common-themes/1313
loop8 7:8 0 105.2M 1 loop /snap/core/17292
sda 8:0 0 20G 0 disk
└─sda1 8:1 0 20G 0 part /
sdb 8:16 0 1G 0 disk
sr0 11:0 1 1024M 0 rom
dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ sudo fdisk /dev/sdb
[sudo] password for dhilip:

Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x73786a80.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):
```

```
Using default response p.
Partition number (1-4, default 1):
First sector (2048-2097151, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-2097151, default 2097151):

Created a new partition 1 of type 'Linux' and of size 1023 MiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 261888 4k blocks and 65536 inodes
Filesystem UUID: 9bdc26bb-5eaf-4499-9189-943da8ea1634
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

dhilip@ubuntu:~$
```

```

dhilip@ubuntu:~$ sudo mkdir /mnt/virtualdisk
dhilip@ubuntu:~$ sudo mount /dev/sdb1 /mnt/virtualdisk
dhilip@ubuntu:~$ df -h

```

| Filesystem  | Size | Used | Avail | Use% | Mounted on                     |
|-------------|------|------|-------|------|--------------------------------|
| udev        | 960M | 0    | 960M  | 0%   | /dev                           |
| tmpfs       | 197M | 1.8M | 195M  | 1%   | /run                           |
| /dev/sda1   | 20G  | 7.1G | 12G   | 39%  | /                              |
| tmpfs       | 984M | 0    | 984M  | 0%   | /dev/shm                       |
| tmpfs       | 5.0M | 4.0K | 5.0M  | 1%   | /run/lock                      |
| tmpfs       | 984M | 0    | 984M  | 0%   | /sys/fs/cgroup                 |
| /dev/loop0  | 1.0M | 1.0M | 0     | 100% | /snap/gnome-logs/61            |
| /dev/loop1  | 55M  | 55M  | 0     | 100% | /snap/core18/1066              |
| /dev/loop2  | 15M  | 15M  | 0     | 100% | /snap/gnome-characters/296     |
| /dev/loop3  | 3.8M | 3.8M | 0     | 100% | /snap/gnome-system-monitor/100 |
| /dev/loop4  | 89M  | 89M  | 0     | 100% | /snap/core/7270                |
| /dev/loop5  | 150M | 150M | 0     | 100% | /snap/gnome-3-28-1804/67       |
| /dev/loop6  | 4.2M | 4.2M | 0     | 100% | /snap/gnome-calculator/406     |
| /dev/loop7  | 43M  | 43M  | 0     | 100% | /snap/gtk-common-themes/1313   |
| tmpfs       | 197M | 16K  | 197M  | 1%   | /run/user/121                  |
| tmpfs       | 197M | 24K  | 197M  | 1%   | /run/user/1000                 |
| /dev/loop8  | 106M | 106M | 0     | 100% | /snap/core/17292               |
| /dev/loop9  | 67M  | 67M  | 0     | 100% | /snap/core24/1643              |
| /dev/loop10 | 128K | 128K | 0     | 100% | /snap/bare/5                   |
| /dev/loop11 | 896K | 896K | 0     | 100% | /snap/gnome-logs/129           |
| /dev/loop12 | 1.8M | 1.8M | 0     | 100% | /snap/gnome-system-monitor/193 |
| /dev/loop13 | 9.5M | 9.5M | 0     | 100% | /snap/gnome-characters/805     |
| /dev/loop14 | 2.2M | 2.2M | 0     | 100% | /snap/gnome-calculator/972     |
| /dev/loop15 | 92M  | 92M  | 0     | 100% | /snap/gtk-common-themes/1535   |
| /dev/loop16 | 165M | 165M | 0     | 100% | /snap/gnome-3-28-1804/198      |
| /dev/loop17 | 402M | 402M | 0     | 100% | /snap/mesa-2404/1839           |
| /dev/sdb1   | 991M | 2.6M | 922M  | 1%   | /mnt/virtualdisk               |

```

dhilip@ubuntu:~$

```

```

dhilip@ubuntu:~$ echo "Virtual Block Test Data" | sudo tee /mnt/virtualdisk/test_data.txt
Virtual Block Test Data

```

```

dhilip@ubuntu:~$ cat /mnt/virtualdisk/test_data.txt
Virtual Block Test Data
dhilip@ubuntu:~$

```

```

dhilip@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
loop0 7:0 0 1.7M 1 loop /snap/gnome-system-monitor/193
loop1 7:1 0 91.7M 1 loop /snap/gtk-common-themes/1535
loop2 7:2 0 4M 1 loop /snap/gnome-calculator/406
loop3 7:3 0 2.1M 1 loop /snap/gnome-calculator/972
loop4 7:4 0 149.9M 1 loop /snap/gnome-3-28-1804/67
loop5 7:5 0 105.2M 1 loop /snap/core/17292
loop6 7:6 0 54.4M 1 loop /snap/core18/1066
loop7 7:7 0 3.7M 1 loop /snap/gnome-system-monitor/100
loop8 7:8 0 66.8M 1 loop /snap/core24/1643
loop9 7:9 0 14.8M 1 loop /snap/gnome-characters/296
loop10 7:10 0 614.5M 1 loop /snap/gnome-46-2404/164
loop11 7:11 0 402M 1 loop /snap/mesa-2404/1839
loop12 7:12 0 164.8M 1 loop /snap/gnome-3-28-1804/198
loop13 7:13 0 1008K 1 loop /snap/gnome-logs/61
loop14 7:14 0 9.4M 1 loop /snap/gnome-characters/805
loop15 7:15 0 4K 1 loop /snap/bare/5
loop16 7:16 0 42.8M 1 loop /snap/gtk-common-themes/1313
loop17 7:17 0 868K 1 loop /snap/gnome-logs/129
sda 8:0 0 20G 0 disk
l-sda1 8:1 0 20G 0 part /
sdb 8:16 0 1G 0 disk
l-sdb1 8:17 0 1023M 0 part
sr0 11:0 1 1024M 0 rom

```

```
dhilip@ubuntu:~$ sudo mount /dev/sdb1 /mnt/virtualdisk
[sudo] password for dhilip:
dhilip@ubuntu:~$ cat /mnt/virtualdisk/test__data.txt
Virtual Block Test Data
dhilip@ubuntu:~$
```

**24. To showcase the virtual machine migration based on the certain condition from one host to the other.**

**Aim:**

To demonstrate the migration of a virtual machine from one host/storage location to another using VMware Workstation Pro and verify the migrated virtual machine by executing basic Linux commands.

**Procedure:**

1. Power off the Ubuntu VM properly from Ubuntu.
2. Close VMware Workstation Pro.
3. Open File Explorer and navigate to:

C:\Users\ADMIN\Documents\Virtual Machines

4. Locate the original VM folder:

Ubuntu-VM

5. Copy the complete Ubuntu-VM folder.
6. Paste it in the same location and rename the copied folder as:

Ubuntu-VM-Migrated

7. Open VMware Workstation Pro.
8. Select:  
File → Open
9. Navigate to:

C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM-Migrated

10. Select the file with type:  
VMware virtual machine configuration (.vmx).
11. Click Open.
12. VMware displays:  
“Did you move or copy this virtual machine?”
13. Select:  
I Copied It
14. Click OK.

15. Power on the migrated Ubuntu virtual machine.

16. After Ubuntu starts, open the Terminal using:

Ctrl + Alt + T

17. Verify the migrated VM using:

hostname

18. Check the Linux system information using:

uname -a

19. If the Ubuntu VM starts successfully and the commands execute correctly, the migration is successfully demonstrated.

## Implementation:

